

1 Problem setting

A scalar advection-diffusion problem without reaction.

$$u_t + \nabla \cdot (F(u) - D(u)\nabla u) = 0, \mathbf{x} \in \mathbb{R}^d \quad (1)$$

The variable $D(u)$ will be treated with Kirchhoff transformation. The diffusion coefficient D is treated as a constant here. This will not affect the following discussion on numerical scheme.

$$u_t + \nabla \cdot F(u) - D \Delta u = 0, \mathbf{x} \in \mathbb{R}^d \quad (2)$$

1.1 Finite volume scheme

Replace scalar u with cell averaged value

$$\bar{u} = \frac{1}{|E|} \int_E u(\mathbf{x}, t) d\mathbf{x} \quad (3)$$

and Lax-Friedrichs numerical flux

$$f(u) = \frac{1}{2} [F(u^+) + F(u^-) \cdot \mathbf{v}_E - \alpha(u^+ - u^-)] \quad (4)$$

The finite volume scheme of advection-diffusion problem is:

$$\bar{u}_t + \frac{1}{|E|} \int_{\partial E} (\hat{f}(\bar{u}) - \nabla \bar{u} \cdot \mathbf{v}_E) d\sigma(\mathbf{x}) = 0 \quad (5)$$

In 2D quadrilateral logically rectangular mesh, we apply gaussian quadrature every edge as the approximation of integral of flux. I use Multi-level WENO reconstruction scheme to reconstruct the point value in advection numerical flux.

$$\hat{f}(u) = \frac{1}{2} [F(\mathcal{R}^+(\{\bar{u}\}_+)) + F(\mathcal{R}^-(\{\bar{u}\}_-)) \cdot \mathbf{v}_E - \alpha(\mathcal{R}^+(\{\bar{u}\}_+) - \mathcal{R}^-(\{\bar{u}\}_-))] \quad (6)$$

Let $\{\bar{u}\}_+ = \{\bar{u}_i \mid i \in \text{WENO stencil } I_+\}$. Diffusion flux is different from its advection counterpart. I reconstruct derivative across the edge by sampling along the normal direction.

2 Computation of Jacobian

Implicit method contains solving a non linear system and a linear subproblem. I write the general non linear system as:

$$\bar{u}_t + \mathcal{F}(\bar{u}) - \mathcal{G}(\bar{u}) = 0 \quad (7)$$

where $\mathcal{F}(\bar{u})$ represents advection part, and $\mathcal{G}(\bar{u})$ represents diffusion part. The semi discretized form is

$$\bar{u}^{n+1} - \bar{u}^n + \Delta t [\mathcal{F}(\bar{u}^{n+1}) - \mathcal{G}(\bar{u}^{n+1})] = 0 \quad (8)$$

There are many well-researched numerical methods that can be used to solve such non linear system.

2.1 ML-WENO reconstruction and derivatives

I use ML-WENO reconstruction scheme to reconstruct point values from cell-averaged value... I use two stage non linear weighting. The first stage is calculated as

$$\check{\omega}_l = \frac{\omega_l}{(\sigma_l + \varepsilon h_0^2)^{\eta_l}} \quad \eta_l = \begin{cases} 1, & \text{if } r = 1 \\ 3, & \text{if } r = 2 \\ 4, & \text{if } r \geq 3 \end{cases} \quad (9)$$

$$\bar{\omega}_l = \frac{\check{\omega}_l}{\sum_k \check{\omega}_k} \quad (10)$$

The the second stage is calculated based on first stage:

$$\hat{\omega}_l = \frac{\bar{\omega}_l}{(\sigma_l + \varepsilon h_0^2)^{sr_l}} \quad (11)$$

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_k \hat{\omega}_k} \quad (12)$$

The reconstruction is expressed as the weighted linear combination of stencil polynomials.

$$\mathcal{R}(\mathbf{x}) = \sum_{l \in L} P_l(\mathbf{x}) \tilde{\omega}_l(\bar{u}) \quad (13)$$

Suppose the reconstruction happens in a given stencil I . The cell averaged values $\bar{u}$ in this stencil is represented as $\{\bar{u}_s\}_{s \in S}$. The derivative of this reconstruction with respect to $\bar{u}_s$ is:

$$\frac{\partial \mathcal{R}}{\partial \bar{u}_s} = \sum_{l \in L} \left(\frac{\partial P_l}{\partial \bar{u}_s} \tilde{\omega}_l + P_l \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} \right) \quad (14)$$

It consists of two parts, derivative of stencil polynomials $\frac{\partial P_l}{\partial \bar{u}_s}$ and derivative of non linear weights $\frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s}$. The stencil polynomial is a combination of single polynomials satisfying

$$\frac{1}{|E_k|} \int_{E_k} p_s(\mathbf{x}) d\mathbf{x} = \begin{cases} 1 & s = k \\ 0 & s \neq k \end{cases} \quad (15)$$

I use tensorproduct polynomials for logically rectangular mesh. The tensorproduct polynomial is written in the following form. Coefficients c_α are determined by solving a linear system satisfying (15).

$$p_s = \sum_{i < I} a_i^s \left(\frac{x - x_S}{h_S} \right)^i \sum_{j < J} b_j^s \left(\frac{y - y_S}{h_S} \right)^j = \sum_{|\alpha| \leq (I-1) \times (J-1)} \tilde{c}_\alpha^s \left(\frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha \quad (16)$$

The stencil polynomial is expressed in terms of a combination of all these polynomials.

$$P_l(\mathbf{x}) = \sum_{s \in S} p_s^l(\mathbf{x}) \bar{u}_s \quad (17)$$

Combining cell averaged values $\bar{u}_s$ with coefficients

$$c_\alpha = \sum_{s \in \mathcal{S}} \tilde{c}_\alpha^s \bar{u}_s \quad (18)$$

The derivative of polynomial is straightforward. The derivative of non linear weight is a little more complicated. Equation 14 can be rewritten as:

$$\frac{\partial \mathcal{R}}{\partial \bar{u}_s} = \sum_{l \in L} p_s^l(\mathbf{x}) \tilde{\omega}_l + \sum_{l \in L} P_l \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} \quad (19)$$

The derivatives of non linear weights are:

$$\begin{aligned} \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} &= \frac{\hat{\omega}_{l,s}}{\sum_k \hat{\omega}_k} - \frac{\hat{\omega}_l \sum_k \hat{\omega}_{k,s}}{(\sum_k \hat{\omega}_k)^2} \\ \frac{\partial \hat{\omega}_l}{\partial \bar{u}_s} &= \frac{\bar{\omega}_{l,s}}{(\sigma_l + \varepsilon h_0^2)^{sr_l}} - sr_l \frac{\bar{\omega}_l \sigma_{l,s}}{(\sigma_l + \varepsilon h_0^2)^{sr_l+1}} \\ \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} &= \frac{\hat{\omega}_{l,s}}{\sum_k \hat{\omega}_k} - \frac{\tilde{\omega}_l \sum_k \tilde{\omega}_{k,s}}{(\sum_k \tilde{\omega}_k)^2} \\ \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} &= -\eta_l \sigma_{l,s} \frac{\omega_l}{(\sigma_l + \varepsilon h_0^2)^{\eta_l+1}} \end{aligned} \quad (20)$$

We calculate the derivatives of non linear weights with chain rule. We also need the derivative of smoothness indicator. The Jiang-Shu smoothness indicator is:

$$\sigma_{JS} = \sum_{0 < |\beta| < r} \frac{h_0^{2|\beta|}}{|E_0|} \int_{E_0} \left(\sum_{|\alpha| < r} c_\alpha \mathcal{D}^\beta \left(\frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha \right)^2 d\mathbf{x} \quad (21)$$

where order $r = (I - 1) \times (J - 1)$, a constant determined by the size of stencil. We present an approximation of the Jiang-Shu smoothness indicator (cite) by omitting all cross product terms.

$$\begin{aligned} \tilde{\sigma}_P &= \sum_{0 < |\alpha| < r} \sum_{0 < \beta \leq \alpha} \left(\frac{h_0}{h_S} \right)^{2|\beta|} c_\alpha^2 \left(\frac{\alpha!}{(\alpha - \beta)!} \right)^2 \frac{1}{|E_0|} \int_{E_0} \left(\frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^{2(\alpha - \beta)} d\mathbf{x} \\ &= \sum_{0 < |\alpha| < r} \sum_{0 < \beta \leq \alpha} \mathcal{C}(\alpha, \beta) c_\alpha^2 \end{aligned} \quad (22)$$

the derivative of $\tilde{\sigma}_P$ can be written as:

$$\begin{aligned} \frac{\partial \tilde{\sigma}_P}{\partial \bar{u}_s} &= \sum_{0 < |\alpha| < r} \sum_{0 < \beta \leq \alpha} \mathcal{C}(\alpha, \beta) 2c_\alpha \frac{\partial c_\alpha}{\partial \bar{u}_s} \\ \frac{\partial c_\alpha}{\partial \bar{u}_s} &= \tilde{c}_\alpha^s \end{aligned} \quad (23)$$

2.2 Derivative of flux

Implicit method requires calculation of derivatives of both advection and diffusion flux.

2.2.1 Derivative of advection flux

Differentiate equation 6 is straight forward.

$$\begin{aligned} \frac{\partial \hat{f}(u)}{\partial \bar{u}_s} &= \frac{1}{2} \left[\left(\frac{\partial \mathcal{R}^+}{\partial \bar{u}_s} + \frac{\partial \mathcal{R}^-}{\partial \bar{u}_s} \right) \frac{\partial F}{\partial \bar{u}_s} \cdot \mathbf{v}_E - \alpha \left(\frac{\partial \mathcal{R}^+}{\partial \bar{u}_s} - \frac{\partial \mathcal{R}^-}{\partial \bar{u}_s} \right) \right] \\ &= \frac{1}{2} \left[\frac{\partial \mathcal{R}^+}{\partial \bar{u}_s} \left(\frac{\partial F}{\partial \bar{u}_s} \cdot \mathbf{v}_E - \alpha \right) + \frac{\partial \mathcal{R}^-}{\partial \bar{u}_s} \left(\frac{\partial F}{\partial \bar{u}_s} \cdot \mathbf{v}_E + \alpha \right) \right] \end{aligned} \quad (24)$$

2.2.2 Derivative of diffusion flux

3 Implicit time stepping

3.1 Backward Euler

3.2 Implicit SSP time stepping

4 Parallelism

Parallel solver is inevitable in large scale computation. We talk about mesh partitioning strategy across different processors in the following sections.

4.1 Logically rectangular mesh and stencils

Cells in logically rectangular mesh maintains logical order. Logically rectangular mesh shows more flexibility in some computation scenarios than cartesian mesh. Figure 1 shows a full view of a logically rectangular (quadrilateral) mesh. Counting all

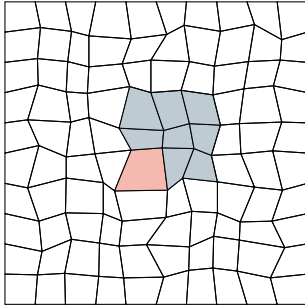


Fig. 1 Logically rectangular mesh displayed in full. The index of a stencil (colored cells) is the same as the index of the red cell.

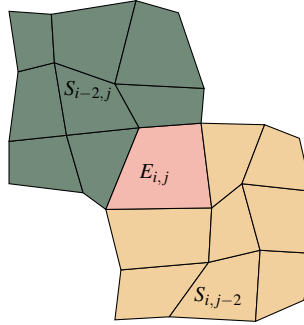


Fig. 2 Two sample stencils used by cell $E_{i,j}$. The index of these two stencils are $S_{i-2,j}$ and $S_{i,j-2}$. Consistent with the index of bottom left cell.

stencils without repeating is convenient in the logically rectangular mesh setting. We

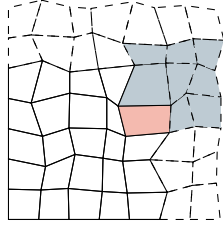


Fig. 3 Illustration of mesh partition. Ghost cells are drawn with dashed lines. The size of ghost layer is determined by the number of cells need in the time stepping computation.

use the index of bottom left cell as the index of the corresponding stencil. A $M \times N$ mesh will then contain $(M - I + 1) \times (N - J + 1)$ stencils of size $I \times J$.

4.2 Mesh partition

I examine the following partition of mesh without lossing generality. The dashed line indicates ghost cells distributed by MPI communication. The size of ghost layer is determined by the largest stencil size required for cells in the ghost layer. Increasing the order of reconstruction (increasing the size of corresponding stencil) will thus increase the communication cost. However, the distribution of mesh only occur once in the beginning of the computation. Mesh will not change during the process of compuation. Figure 4.2 shows a stencil covers ghost layer.

4.3 Sparsity pattern of Jacobian matrix

4.4 Optimal preconditioner

- 1. Incomplete LU decomposition
- 2. Additive schwarz

4.5 Scalability